NIMBUS-9¾

HARDWARE FULL-STACK — DRAWING PACKAGE

Real Quidditch · broom + three balls · BOM · OpenSCAD CAD · drawings

SHEET 1	BROOM	manned eVTOL, 8 internal ducted fans
SHEET 2	GOLDEN SNITCH	evasion drone, real flapping wings
SHEET 3	BLUDGER	pursuit + soft-tag, zero contact
SHEET 4	QUAFFLE	catch / carry / throw / score
SHEET 5	MATCH FLEET	rollup cost for one 7v7 match

"Magic? No. We just wrote the collision-avoidance really well."

generated 2026-06-26 · geometry true to nimbus_fc params · v0.1

NIMBUS-9¾ BROOM

Manned eVTOL — 1 rider, 8 ducted fans, fly-by-intent

Project NIMBUS-9¾ · Real Quidditch · hardware full-stack

ISOMETRIC

SIDE

TOP

CUTAWAY — internal drive revealed

KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Overall length `core/params.py LEN = 2.4 m`	**2400 mm**
Lift fans `_default_fan_layout (exact x/y)`	**8 × EDF (internal)**
Thrust-to-weight `analysis/flight_thermal.py`	**2.45 (hover @ 41%)**
All-up mass `rider ~80 + airframe ~40`	**120 kg**
Hover endurance `2600 Wh — a sprinter, pit doctrine`	**~89 s / charge**
Cooling margin `lift air = coolant + heat pipes`	**17× waste heat**
Soft fence `fence_ceiling / fence_floor`	**ceiling 18 m / floor 3 m**
Control core `rust/nimbus_core + firmware/hal.py`	**Cortex-M4F @ 400 Hz**

NOTE

A bare stick has no disc area, so it's a ~90 s sprinter (hence the F1-style hot-swap pits the software already enforces). Fans are fully internal — prop exposure 0; the lift air doubles as motor coolant, with heat pipes spreading the rest over the 2.4 m shaft. Budget closes: FLIGHT + THERMAL both PASS.

BILL OF MATERIALS — BRM (31 line items, prototype qty 1)

REF	PART	SPEC	QTY	$ EA	$ EXT
BRM-01	Carbon-fibre monocoque body p...	2.4 m, hides the fan bay; SI->mm x1000 in cad/broom.scad	1	2,200	2,200
BRM-02	Handle shaft + brass pommel	tapered Ø34->Ø22 mm, leather-wrapped grip	1	180	180
BRM-03	Bristle fairing leaf	~780 carbon-straw strands, aerodynamic shroud	1	640	640
BRM-04	Electric ducted fan (EDF) unit	360 N max each; main-lift in bristle shroud + in-shaft trim fans	8	520	4,160
BRM-05	Brushless motor for EDF	~12 kW peak, 6S-14S, duct-matched	8	340	2,720
BRM-06	ESC (electronic speed control...	200 A, telemetry, 1st-order ~40 ms response	8	150	1,200
BRM-07	Li-ion battery pack	2600 Wh usable, hot-swappable cartridge	1	3,100	3,100
BRM-08	Battery management system (BM...	14S smart BMS, cell balancing + SoC telemetry	1	260	260
BRM-09	Power distribution board	800 A bus, redundant rails	1	180	180
BRM-10	Flight controller MCU	Cortex-M4F (thumbv7em-none-eabihf), 400 Hz loop	1	95	95
BRM-11	IMU (redundant)	triple-redundant 6-DOF, 1 kHz	3	55	165
BRM-12	Magnetometer	3-axis, maneuver-compensated	2	18	36
BRM-13	Barometer	high-res, for altitude hold	2	12	24
BRM-14	RTK-GPS receiver + antenna	cm-grade, dual-band	1	420	420
BRM-15	UWB ranging tag	indoor cm positioning to pitch anchors	2	40	80
BRM-16	Solid-state LiDAR	360deg, edge collision avoidance	2	600	1,200
BRM-17	Low-latency datalink radio	redundant 5G + dedicated mesh	2	140	280
BRM-18	Ballistic recovery parachute ...	low-altitude rated, pyro deploy	1	1,500	1,500
BRM-19	Airbag skirt	underbody impact airbag, impact-triggered	1	450	450
BRM-20	Master kill-switch receiver	referee e-stop -> gentle descent	1	60	60
BRM-21	5-point harness + saddle	straddle seat, energy-absorbing	1	320	320
BRM-22	Rider intent controls	handlebar grips, throttle, lean sensors	1	210	210
BRM-23	Brass compass/attitude gauge	cosmetic + real attitude readout	1	90	90
BRM-24	LED trajectory strip	addressable, night-match light show	1	70	70
BRM-25	Heat pipe (sintered copper)	Ø6 mm, full-shaft length, motor->shaft spreader	4	22	88
BRM-26	Air-over-motor duct shroud	routes lift airflow across windings (air = coolant)	8	16	128
BRM-27	In-duct winding temp sensor	NTC, one per fan, feeds the governor	8	3	24
BRM-28	Thermal interface material	phase-change pad, motor + ESC to spreader	1	18	18
BRM-29	DShot ESC wiring harness	bidirectional DShot600, 8 channels + telem	1	45	45
BRM-30	CAN-FD vehicle bus	5 Mbit, BMS + ESC telem + pose to Dementor	1	30	30
BRM-31	Hardware abstraction layer	firmware skin around the Rust core	1	0	0

UNIT TOTAL (prototype): **$19,973**

DRAWN	DATE	UNITS	SCALE	PART	REV	CAD
NIMBUS HW	**2026-06-26**	**mm / SI**	**NTS**	**BRM**	**v0.1**	**cad/broom.scad**

GOLDEN SNITCH

Walnut-sized evasion drone with real flapping wings

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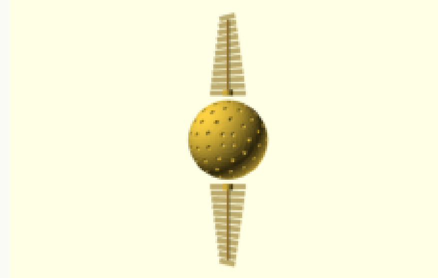
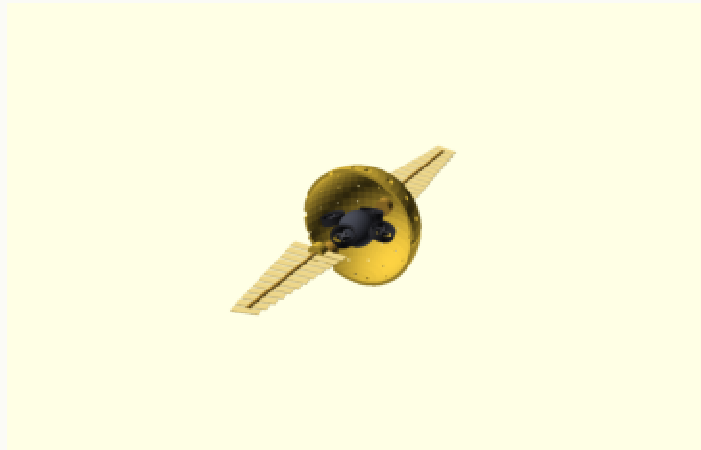
ISOMETRIC

SIDE

TOP



CUTAWAY — internal drive revealed



KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Shell radius `ball/params.py snitch.radius = 0.04 m`	**40 mm (Ø80)**
Mass target `snitch.mass = 0.05 kg`	**50 g**
Top speed `snitch.max_speed`	**19 m/s**
Max accel `snitch.max_accel (absurdly agile)`	**32 m/s²**
Lift `swarm core — the real flight`	**4 × micro-EDF (internal)**
Wings `coreless gearmotor driven`	**2 × flapping, 8–14 Hz**
Capture `capture_radius 0.22 / capture_dwell`	**hand held 0.25 s**
Danger sense `snitch.danger_radius`	**4.0 m**

NOTE
OFFICIAL STORY: it flies by flapping its wings. The wings are thin and genuinely flutter (SNT-02/03/04). The lift is actually the internal micro-EDF swarm (SNT-05). Do not explain this to Potterheads.

BILL OF MATERIALS — SNT (14 line items, prototype qty 1)

REF	PART	SPEC	QTY	$ EA	$ EXT
SNT-01	Gold-clad vented sphere shell	Ø80 mm (R=40 mm), walnut-sized, dimple vents	1	90	90
SNT-02	Animatronic flapping wing ass...	thin filigree vane on hinge, span ~76 mm	2	28	56
SNT-03	Coreless gearmotor + crank	6 mm coreless, flap linkage, 8-14 Hz beat	2	9	18
SNT-04	Wing membrane	5 micron gold-tint mylar, feather-cut	2	4	8
SNT-05	Micro ducted-fan lift unit	Ø16 mm internal, the ACTUAL lift	4	22	88
SNT-06	Micro brushless motor	4 mm, paired to micro-EDF	4	11	44
SNT-07	Micro ESC (4-in-1)	5 A, integrated	1	26	26
SNT-08	Nano flight controller	STM32 class, IMU on-board, predictive evasion	1	38	38
SNT-09	Optical-flow + ToF sensor	360deg-ish reach detection	2	16	32
SNT-10	Micro LiDAR / proximity ring	detects incoming hands/airframes	1	45	45
SNT-11	Capacitive capture sensor	registers an enclosing hand	1	12	12
SNT-12	Micro Li-Po cell	1S 300 mAh, lightweight	1	8	8
SNT-13	Gold micro-LED	glow so the crowd can track it	1	5	5
SNT-14	2.4 GHz telemetry link	reports pose to the Dementor referee	1	14	14

UNIT TOTAL (prototype): **$484**

DRAWN	DATE	UNITS	SCALE	PART	REV	CAD
NIMBUS HW	**2026-06-26**	**mm / SI**	**NTS**	**SNT**	**v0.1**	**cad/snitch.scad**

BLUDGER

The skull-cracker, with the skull-cracking removed

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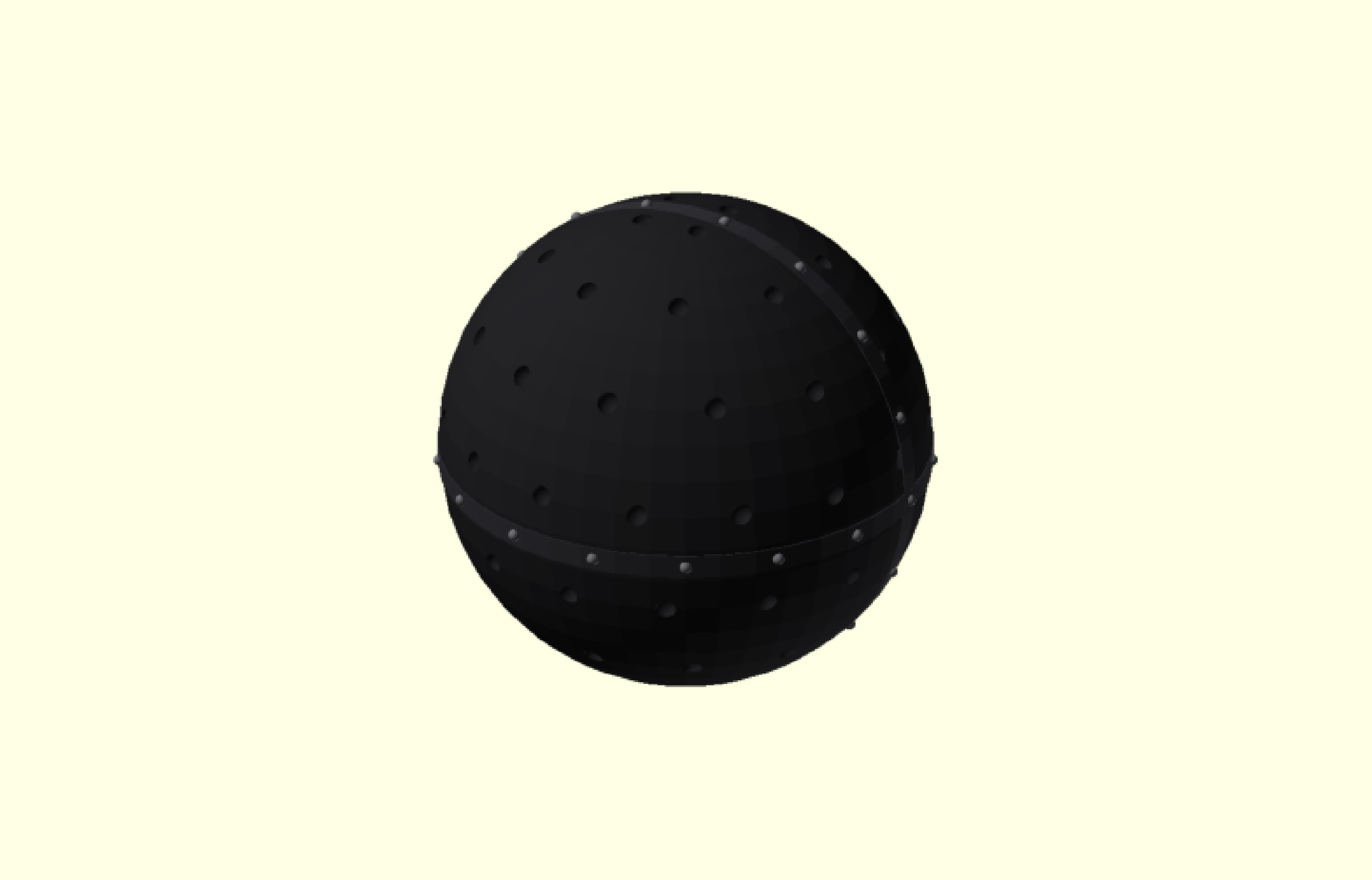
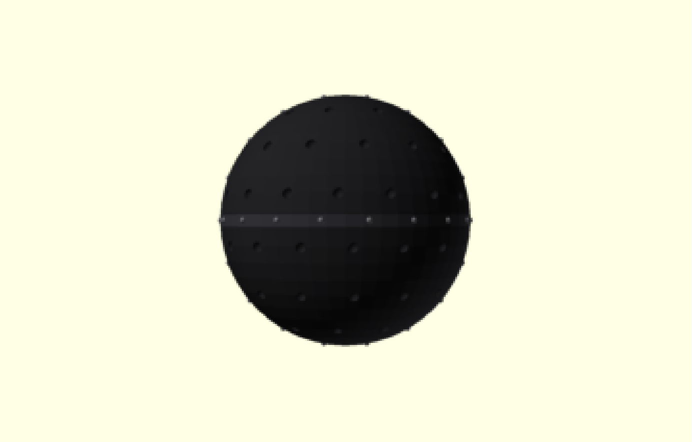
ISOMETRIC



SIDE



TOP

CUTAWAY — internal drive revealed

KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Shell radius `ball/params.py bludger.radius = 0.15 m`	**150 mm (Ø300)**
Mass `bludger.mass — looks iron, isn't`	**1.2 kg (foam)**
Top speed `bludger.max_speed`	**15 m/s**
Max accel `bludger.max_accel (most aggressive)`	**22 m/s²**
Lift `swarm core`	**4 × EDF (internal)**
Soft-tag radius `tag_radius (outside no-contact floor)`	**1.5 m**
Bat deflect `bat_radius`	**1.6 m**
Contact `ball/avoidance.py, worst clr +0.40 m`	**0 — software hit**

NOTE
All threat, no contact. The 'hit' is a proximity tag registered in software ~1.5 m out; the foam shell never has to reach a person.

BILL OF MATERIALS — BLG (12 line items, prototype qty 1)

REF	PART	SPEC	QTY	$ EA	$ EXT
BLG-01	Black impact-foam sphere shell	Ø300 mm (R=150 mm), riveted-iron look	1	70	70
BLG-02	Equator rivet bands (cosmetic)	two crossed bands + dummy rivets	1	18	18
BLG-03	Ducted-fan lift unit	Ø40 mm internal, internal swarm	4	40	160
BLG-04	Brushless motor	per fan, high power-to-weight	4	22	88
BLG-05	4-in-1 ESC	20 A, telemetry	1	38	38
BLG-06	Flight controller	IMU on-board, lead-pursuit guidance	1	42	42
BLG-07	LiDAR / proximity sensor	target ranging for the soft tag	2	55	110
BLG-08	Bat-swing detector	sees the beater's marked bat	1	16	16
BLG-09	No-contact avoidance compute	clamps velocity to a stop at the shell	1	0	0
BLG-10	Li-Po pack	3S, sized for sprint pursuits	1	24	24
BLG-11	Status LED	angry red glow	1	5	5
BLG-12	Telemetry link	reports pose to the Dementor	1	14	14

UNIT TOTAL (prototype): $585

DRAWN	DATE	UNITS	SCALE	PART	REV	CAD
NIMBUS HW	2026-06-26	mm / SI	NTS	BLG	v0.1	cad/bludger.scad

QUAFFLE

The friendly one — catch, carry, throw, score
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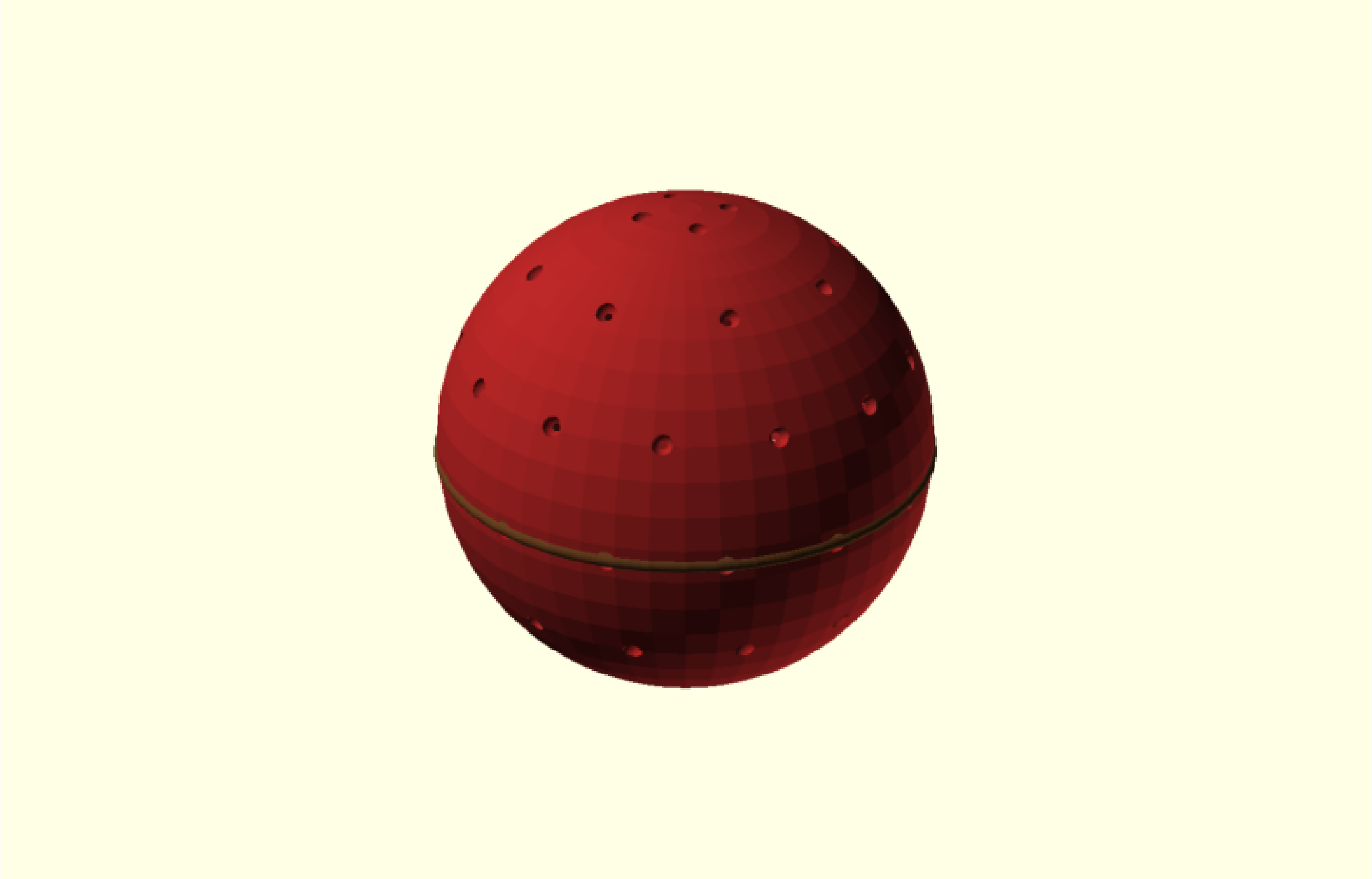
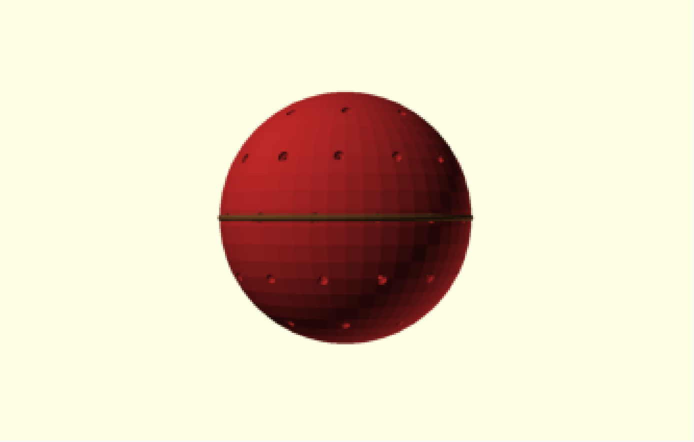
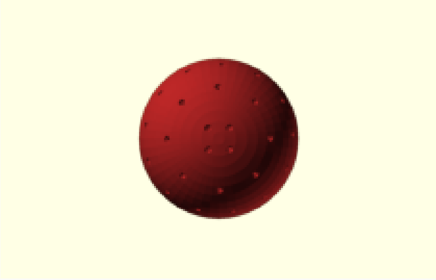
ISOMETRIC



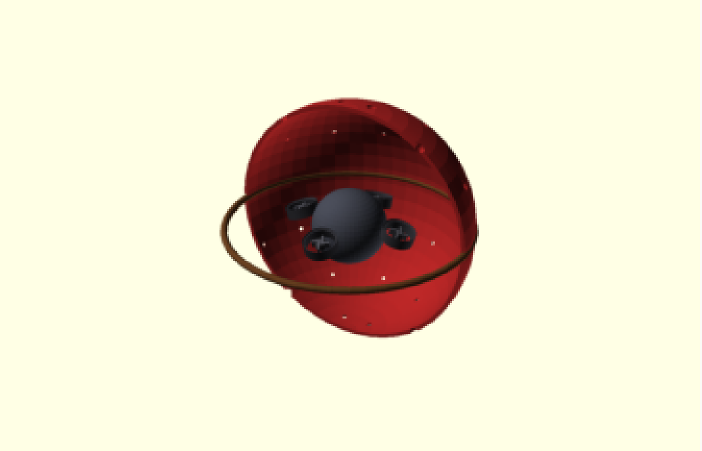
SIDE



TOP



CUTAWAY — internal drive revealed



KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Shell radius `ball/params.py quaffle.radius = 0.18 m`	**180 mm (Ø360)**
Mass `quaffle.mass — light despite its size`	**0.5 kg**
Top speed `quaffle.max_speed (gentle)`	**7 m/s**
Catch radius `quaffle.catch_radius (forgiving)`	**0.45 m**
Lift `swarm core, soft hover`	**4 × EDF (internal)**
On catch `ball/quaffle.py capacitive sense`	**thrust → 0 in hand**
Scoring `auto pass-through detection`	**hoop optical gate**
Safety `even the nice ball won't bonk you`	**no-contact floor**

NOTE
Soft hover, generous catch, rides with its holder, then throws to the nearest hoop with tunable aim-assist (on for youth, off for pros).

BILL OF MATERIALS — QAF (12 line items, prototype qty 1)

REF	PART	SPEC	QTY	$ EA	$ EXT
QAF-01	Red grippy foam sphere shell	Ø360 mm (R=180 mm), seamed + finger grooves	1	75	75
QAF-02	Grab-handle ring	recessed leather ring, holds the catch sensor	1	14	14
QAF-03	Ducted-fan lift unit	Ø48 mm internal, internal swarm	4	42	168
QAF-04	Brushless motor	low-speed, smooth	4	20	80
QAF-05	4-in-1 ESC	15 A, telemetry	1	34	34
QAF-06	Flight controller	IMU on-board, throw + aim-assist	1	40	40
QAF-07	Capacitive catch sensor array	detects a holding hand -> thrust to 0	1	18	18
QAF-08	Hoop pass-through beacon	IR tag read by the hoop optical gate	1	10	10
QAF-09	No-contact avoidance compute	even the nice ball won't bonk you	1	0	0
QAF-10	Li-Po pack	3S, long hover endurance	1	22	22
QAF-11	Status LED	warm white possession glow	1	5	5
QAF-12	Telemetry link	reports pose to the Dementor	1	14	14

UNIT TOTAL (prototype): **$480**

DRAWN	DATE	UNITS	SCALE	PART	REV	CAD
NIMBUS HW	**2026-06-26**	**mm / SI**	**NTS**	**QAF**	**v0.1**	**cad/quaffle.scad**

ITEM	CODE	MATCH QTY	UNIT $	EXT $
NIMBUS-9¾ BROOM	BRM	14	$19,973	**$279,622**
QUAFFLE	QAF	1	$480	**$480**
BLUDGER	BLG	2	$585	**$1,170**
GOLDEN SNITCH	SNT	1	$484	**$484**
			MATCH TOTAL	$281,756